

Phase 5 – Project Development

1. Introduction

The LegalEase system is developed as an AI-powered legal document generation application. It combines a Streamlit frontend, FastAPI backend, Google Gemini AI, and document formatting modules.

2. Frontend Development

The frontend is developed using **Streamlit**. It provides a simple interface where users can enter:

- Document Type
- Parties involved
- Terms and Conditions
- Effective Date

The interface also provides options to generate the document, preview the content, edit the generated text, and download the final document.

3. Backend Development

The backend is developed using **FastAPI**.

A /generate POST endpoint is used to receive document generation requests. The request contains details such as document type, parties, terms, and dates.

The backend processes the request and sends the required information to the Gemini document generator.

4. AI Integration

Google Gemini is integrated into LegalEase for generating legal document content.

The Gemini document generator receives structured user information and generates the required legal document based on the provided details.

The API key is configured through the .env file for the AI integration.

5. Document Generation

The generated legal content is processed into different document formats.

TXT Generation

The generated content can be exported as a plain text file.

DOCX Generation

The system uses the python-docx library to create an editable Word document. The document can include a logo, headings, terms table, and footer.

PDF Generation

The system uses FPDF to create a formatted PDF containing the generated legal content, logo, headings, bullet-style terms, header, and footer.

HTML Preview

The generated document can be displayed as a styled HTML preview within the Streamlit application.

6. Editing Functionality

After generating the document, users can view the content in an editable area. Users can modify the generated text before downloading the final document.

This provides flexibility to customize the generated legal document according to their requirements.

7. User Interface Integration

The frontend connects the user input, document generation, preview, editing, and download functions into a single workflow.

The basic process is:

Enter Details → Generate → Preview → Edit → Download

8. Development Workflow

The main development workflow of LegalEase is:

User Input → FastAPI Backend → Gemini AI → Generated Content → Formatting → Preview/Edit → Download

The user first provides the required document details. The FastAPI backend processes the request and passes the structured information to the Gemini document generator. The generated content is then formatted and displayed for preview and editing.

9. Security and Configuration

The project uses a .env file for configuring the Gemini API key. This keeps the API configuration separate from the main application code.

The project also considers secure handling of sensitive legal document information.

10. Development Outcome

The development process produces a working LegalEase application with:

- AI-powered legal document generation
- Streamlit-based user interface
- FastAPI backend
- Gemini AI integration
- Editable document preview
- TXT, DOCX, and PDF downloads
- Professional document formatting

11. Conclusion

The Project Development phase brings together the frontend, backend, AI integration, document generation, formatting, preview, editing, and download features to build the LegalEase application.

